

Algebra II with Trigonometry

Chapter 6.3

Olympiad Problems

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Problems

1. Let $f(x) = \sin x + \sin(2x) + \sin(3x)$. What is the maximum value of $f(x)$?
2. Suppose $y = \cos(kx)$ has period $\frac{\pi}{3}$. Without solving, what is the value of k ?
3. For which value(s) of c , is the function $y = \sin(x) + c \sin(x + \pi/3)$ an even function?
4. Let $A = \sin^2\left(\frac{\pi}{8}\right) + \sin^2\left(\frac{3\pi}{8}\right) + \sin^2\left(\frac{5\pi}{8}\right) + \sin^2\left(\frac{7\pi}{8}\right)$. Compute A mentally.
5. Consider $y = 2 \sin x + \sin^2 x$. What is the range of y ?
6. The graphs of $y = a \sin(kx - b)$ and $y = a \cos(kx - b)$ are shifted so that their maximum points coincide for $x > 0$. What must be the distance (in terms of a, k) between their first maximums?